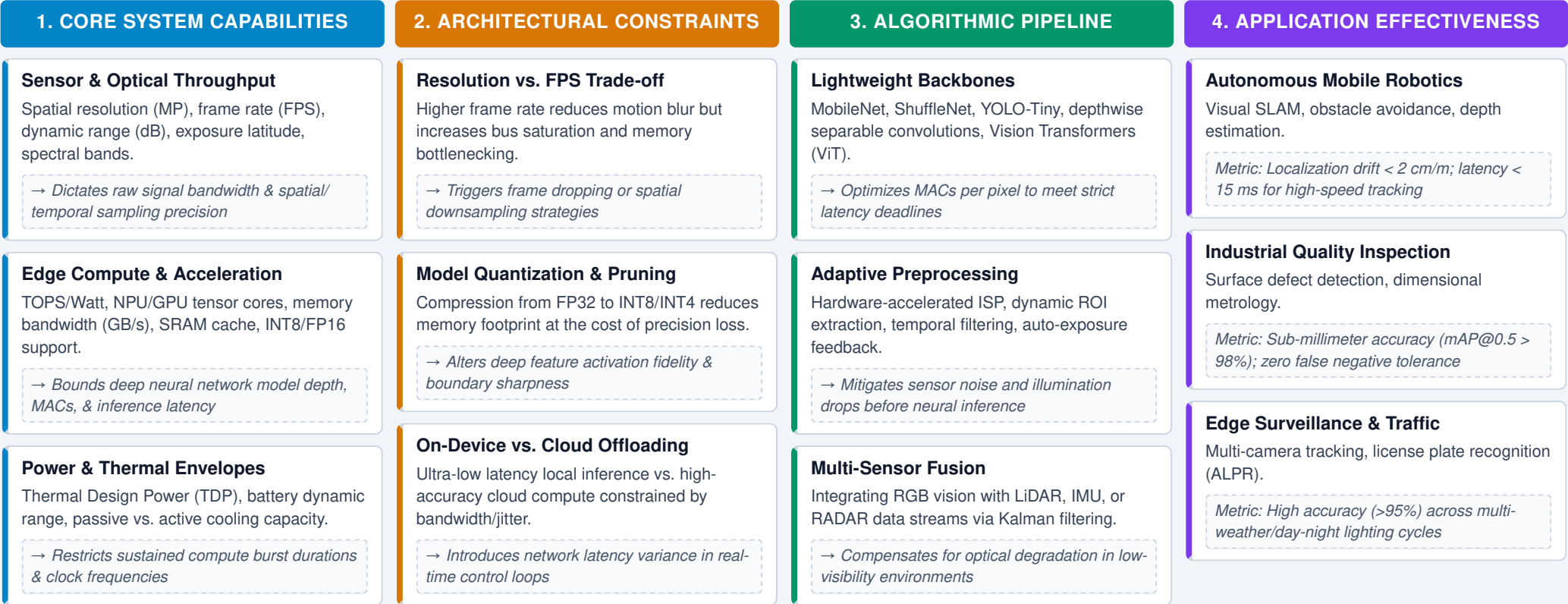


# System Capabilities & Vision-Based Application Effectiveness

An analytical concept map linking hardware, compute, and algorithmic capabilities to edge constraints, selection trade-offs, and downstream task metrics



## Cross-Literature Synthesis & Empirical System Perspectives

| PERSPECTIVE PARADIGM   | SYSTEM CAPABILITY FOCUS               | INTERCONNECTION & DYNAMIC MECHANISM                                                                       | IMPACT ON APPLICATION EFFECTIVENESS                                                                           | KEY LITERATURE REFERENCE                   |
|------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Hardware-Centric       | NPU Tensor Compute & Memory Bandwidth | Memory bandwidth bottlenecking (DRAM vs. SRAM cache) limits CNN layer throughput regardless of peak TOPS. | Restricts real-time 4K video segmentation to <15 FPS unless channel pruning is applied.                       | Sze et al. (2017) / Chen et al. (2020)     |
| Algorithmic Efficiency | Model Quantization (INT8 vs. FP16)    | Post-training quantization reduces model latency by 3x but introduces feature map quantization noise.     | Maintains classification accuracy within 0.8% drop, but causes up to 6.2% mAP drop in fine-grained detection. | Howard et al. (2019) / Jacob et al. (2018) |
| Edge-Cloud Co-Design   | Network Latency & Bandwidth Dynamics  | Adaptive offloading balances local edge compute latency against network transmission jitter.              | Reduces total end-to-end response latency by 42% in collaborative robotic fleets under dynamic network loads. | Teerapittayanon et al. (2017)              |

| PERSPECTIVE PARADIGM  | SYSTEM CAPABILITY FOCUS                 | INTERCONNECTION & DYNAMIC MECHANISM                                                                              | IMPACT ON APPLICATION EFFECTIVENESS                                                                        | KEY LITERATURE REFERENCE                         |
|-----------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Sensor-Compute Fusion | High Dynamic Range (HDR) & Temporal FPS | Event-based cameras paired with neuromorphic processors eliminate motion blur at low power consumption (<100mW). | Enables sub-millisecond reaction times for ultra-high-speed drone collision avoidance in dynamic lighting. | Gallego et al. (2020) / Scaramuzza et al. (2022) |

**Strategic Synthesis:** Selecting vision-based applications requires a balanced co-design between sensor resolution, NPU memory bandwidth, and model complexity. Optimizing software algorithms alone cannot overcome fundamental hardware memory or sensor dynamic range limitations; maximum effectiveness is achieved when hardware throughput and model compression match the exact temporal and spatial demands of the application.